

Rheumatoid Arthritis

AFIAS Anti-CCP Plus

INTENDED USE

AFIAS Anti-CCP Plus is a fluorescence immunoassay (FIA) for the qualitative or semi-quantitative determination of human IgG autoantibodies to cyclic citrullinated peptides (CCP) in human whole blood/serum/plasma. It is useful as an aid in the diagnosis of rheumatoid arthritis (RA) in combination with other clinical and laboratory findings.
For *in vitro* diagnostic use only.

INTRODUCTION

Rheumatoid Arthritis (RA) is a common, systemic autoimmune disease affecting 0.5-1.0 % of the world population. RA is characterized by chronic inflammation of the synovium which can lead to progressive joint destruction, disability and mortality.⁽¹⁾ As joint damage is irreversible, early therapeutic intervention is of paramount importance for the prognosis of patients.^(2,3)

The diagnosis of rheumatological disease are the medical history, clinical findings (including imaging techniques) and serological laboratory tests. Serological diagnostic testing is of growing importance in the early detection and differentiation of RA. The most frequent serological diagnostic testing is the measurement of rheumatoid factor (RF).⁽⁴⁾ The RF antibody is present in about 75% of RA patients, but its specificity is limited, as it is often present in healthy individuals and patients with other rheumatic or inflammatory diseases, autoimmune diseases or chronic infections.⁽⁵⁾

More recently, new specific autoantibodies to citrullinated proteins antigens (ACPAs) have made a crucial contribution to the diagnosis of RA.⁽⁶⁾ Although many assays are available to test for ACPAs to specific antigens, for the clinical management of RA, most ACPA testing is performed using a synthetic cyclic citrullinated protein (CCP) as the antigen to detect ACPAs. An anti-CCP assay is capable to detect the autoantibodies against citrullinated proteins which have a relatively high sensitivity (reportedly between 50-75%) for rheumatoid arthritis and extremely high specificity (about 90%) for RA.⁽⁷⁾ Its high specificity is why the anti-CCP test has become an important part of the diagnostic process for RA.

PRINCIPLE

The test uses a sandwich immunodetection method.
The detector antigens in buffer bind to antibodies in the sample, forming antigen-antibody complexes, and migrate onto nitrocellulose matrix to be captured by the other immobilized-streptavidin on a test strip.
More antibodies in the sample will form more antigen-antibody complexes which lead to stronger fluorescence signal by detector antigens, which is processed by the instrument for AFIAS tests to show Anti-CCP concentration in the sample.

COMPONENTS

- AFIAS Anti-CCP Plus** consists of ‘cartridges’.
- Each sealed aluminum pouch contains two cartridges.
 - Each cartridge packaged in an aluminum pouch has three components including a cartridge part, a detector part and a diluent part.
 - The cartridge part contains the membrane called a test strip which has streptavidin at the test line, and chicken IgY at the control line.
 - The detector part has 2 graules containing anti-human immunoglobulin G-fluorescence conjugate, anti-chicken IgY-fluorescence conjugate, and CCP-biotin conjugate in phosphate buffered saline.
 - The detector diluent part contains sodium azide as a preservative, and tween 20 as a detergent in phosphate buffered saline (PBS).

WARNINGS AND PRECAUTIONS

- For *in vitro* diagnostic use only.
- Follow instructions and procedures described in this ‘Instruction for use’.
- Use only fresh samples and avoid direct sunlight.
- Lot numbers of all the test components (cartridge and ID chip) must match each other.
- Do not interchange the test components between different lots or use the test components after the expiration date, either of which might yield incorrect test result(s).
- Do not reuse cartridges. A cartridge should be used for testing one sample only.
- The cartridge should remain sealed in its original pouch until just before use. Do not use the cartridge, if pouch is damaged or has already been opened.
- Frozen sample should be thawed only once. For shipping, samples must be packed in accordance with local regulations. Sample with severe hemolysis and/or hyperlipidemia must not be used.
- If test components and/or sample are stored in refrigerator, then allow cartridge and sample to be at room temperature for approximately 30 minutes before use.
- The instrument for AFIAS tests may generate slight vibration during use.
- Used cartridges, C-tips and pipette tips should be handled carefully and discarded by an appropriate method in accordance with relevant local regulations.
- The cartridge contains sodium azide (NaN₃), and it may cause certain health issues like convulsions, low blood pressure, low heart rate, loss of consciousness, lung injury and respiratory failure. Avoid contact with skin, eyes, and clothing. In case of contact, rinse immediately with running water.
- No Biotin interference was observed in **AFIAS Anti-CCP Plus** when biotin concentration in the sample was below 200 ng/mL. If a patient has been taking biotin at dosage of more than 0.03 mg a day, it is recommended to test again 24 hours after discontinuation of biotin intake.
- **AFIAS Anti-CCP Plus** will provide accurate and reliable results subject to the below conditions.
 - **AFIAS Anti-CCP Plus** should be used only in conjunction with instrument for AFIAS tests.

- Have to use recommended anticoagulant.
- | Recommended anticoagulant |
|---|
| K ₂ EDTA, Na ₂ EDTA, Sodium citrate |
- **C-tip should be used when the following conditions are met.**
 - C-tip provided with the kit is recommended to obtain correct test result.
 - Whole blood should be immediately tested after collection.
 - Do not perform a test with C-tip on General Mode. It might cause an erroneous result.
 - Excess whole blood around the C-tip should be wiped off.
 - In order to avoid cross-contamination, please do not re-use C-tip for multiple samples.
 - AFIAS cartridge should be inserted and positioned in the cartridge holder prior to the blood sample collection.
 - While collecting blood, be careful not to create air bubbles in the C-tip.

LIMITATIONS OF THE TEST SYSTEM

- The test may yield false positive result(s) due to the cross-reactions and/or non-specific adhesion of certain sample components to the capture/detector antibodies.
- The test may yield false negative result(s) due to the non-responsiveness of the antigens to the antibodies which is the most common if the epitope is masked by some unknown components, so therefore not being able to be detected or captured by the antibodies. The instability or degradation of the antigens with time and/or temperature may also cause false negative result as it makes antigens unrecognizable by the antibodies.
- Other factors may interfere with the test and cause erroneous results, such as technical/procedural errors, degradation of the test components/reagents or presence of interfering substances in the test samples.
- Any clinical diagnosis based on the test result must be supported by a comprehensive judgment of the concerned physician in conjunction with clinical symptoms and other relevant test results.

STORAGE AND STABILITY

Storage condition			
Component	Storage Temperature	Shelf life	Note
Cartridge	2 - 8°C	20 months	Unopened
		1 month	Resealed

- Return an unused cartridge to the spare cartridge zipper bag containing the desiccant pack. Reseal along entire edge of zip-seal.

MATERIALS SUPPLIED

- | | |
|--|----|
| REF SMFP-67 | |
| Components of AFIAS Anti-CCP Plus | |
| ■ Cartridge Box Contains | |
| - Cartridge | 24 |
| - Pipette tip (zipper bag) | 24 |
| - Spare cartridge zipper bag | 1 |
| - ID chip | 1 |
| - Instructions for use | 1 |

MATERIALS REQUIRED BUT SUPPLIED ON DEMAND

Following items can be purchased separately with **AFIAS Anti-CCP Plus**.
Please contact our sales division for more information.

- | | |
|---|--------------|
| ■ Instrument for AFIAS tests | |
| - AFIAS-1 | REF FPRR019 |
| - AFIAS-3 | REF FPRR040 |
| - AFIAS-6 | REF FPRR020 |
| - AFIAS-10 | REF FPRR038 |
| ■ Boditech Anti-CCP Plus Control | REF CFPO-288 |
| ■ C-tip (10 µL) | REF CFPO-91 |

SAMPLE COLLECTION AND PROCESSING

- The sample type for **AFIAS Anti-CCP Plus** is human whole blood/serum/plasma.
- It is recommended to test the sample within 24 hours after collection when collected sample is stored at room temperature.
 - The samples (serum, plasma) should be separated from the clot by centrifugation within 3 hours after the collection of whole blood.
 - The samples (whole blood, serum, plasma) may be stored for a month at 2-8°C prior to being tested. If testing will be delayed more than a month, samples should be frozen at -20°C.
 - The samples (serum, plasma) stored frozen at -20°C for 3 months showed no performance difference.
 - However, the whole blood sample should not be kept in a freezer in any case.
 - As a repeated freeze-thaw cycle may affect the test result, do not refreeze previously frozen samples.
 - Collection of whole blood sample using C-tip
 - ① Hold the C-tip horizontally and touch the surface of the blood with the tip of the C-tip.
 - ② Capillary action will automatically draw the blood sample to C-tip and stop.
 - ③ Wipe off any excess blood around the tip.
 - ④ Double-check if whole blood is fully filled accurately in the C-tip and AFIAS reader is ready for a test on the ‘C-tip mode’.

TEST SETUP

- Check the components of the **AFIAS Anti-CCP Plus** as described below: Cartridges, pipette tips, an ID chip, a spare cartridge zipper bag and an instructions for use.
 - Ensure that the lot number of the cartridge matches that of the ID chip.
 - If the sealed cartridge has been stored in a refrigerator, place them on a clean and flat surface at room temperature for at least 30 minutes before testing.
 - Turn on the instrument for AFIAS tests.
 - Empty the tip box.
 - Insert the ID chip into the ‘ID chip port’.
- ※ Please refer to the instrument for AFIAS tests operation manual for complete information and operating instructions.

TEST PROCEDURE

- **AFIAS-1, AFIAS-3, AFIAS-6**

General mode

1) Insert the cartridge into the cartridge holder.

- 2) Insert a tip into the tip hole of the cartridge.
3) Select the ‘General mode’ in the instrument for AFIAS tests.
4) Take 100 µL of sample (whole blood/ serum/ plasma/ control) using a pipette and dispense it into the sample well of the cartridge.
5) Select the sample type (whole blood or serum/plasma) on the screen.
6) Tap the ‘Start’ button on the screen.
7) The test result will be displayed on the screen after 12 minutes.

C-tip mode

- 1) Insert the cartridge into the cartridge holder.
2) Take 10 µL of whole blood using a C-tip.
3) Insert the whole blood-filled C-tip into the tip hole of the cartridge.
4) Select the ‘C-tip mode’ in the instrument for AFIAS tests.
5) Tap the ‘Start’ button on the screen.
6) The test result will be displayed on the screen after 12 minutes.

► AFIAS-10

Normal mode

- 1) Insert a cartridge into the cartridge holder.
2) Insert a tip into the tip hole of the cartridge.
3) Tap the ‘Load’ button of the bay that holds the cartridge with the tip to read the barcode of the cartridge and please confirm the item name written on the cartridge.
4) Insert the sample tube into the tube rack.
5) Insert the tube rack into the loading part of the sampling station.
6) Select the sample type (whole blood or serum/plasma) on the screen.
7) Tap the ‘Start’ button on the screen.
8) The test result will be displayed on the screen after 12 minutes.

Emergency mode – General tip

- 1) The test procedure is same with the ‘Normal mode 1) – 3)’.
2) Convert the ‘Emergency mode’ in AFIAS-10.
3) Select the tip type (general tip) on the screen.
4) Select the sample type (whole blood or serum/plasma) on the screen.
5) Take 100 µL of the sample using a pipette and dispense it into the sample well of the cartridge.
6) Tap the ‘Start’ button on the screen.
7) The test result will be displayed on the screen after 12 minutes.

Emergency mode – C-tip

- 1) Insert a cartridge into the cartridge holder.
2) Take 10 µL of whole blood using a C-tip.
3) Insert the C-tip with sample into the tip hole of the cartridge.
4) Tap the ‘Load’ button of the bay that holds the cartridge with a tip to read the barcode of the cartridge and please confirm the item name written on the cartridge.
5) Convert the ‘Emergency mode’ in AFIAS-10.
6) Select the tip type (C-tip) on the screen.
7) Tap the ‘Start’ button on the screen.
8) The test result will be displayed on the screen after 12 minutes.

INTERPRETATION OF TEST RESULT

- The instrument for AFIAS tests calculates the test result automatically and displays Anti-CCP level of the test sample in terms of U/mL.

Test result [U/mL]	Display [U/mL]
< 5.0	< 3.2 or value, (Neg)
5.0 ≤, < 300	Value, (Pos)

- Cut-off: 5 U/mL
■ Working range: 3.2 - 300 U/mL

QUALITY CONTROL

- Quality control tests are a part of the good testing practice to confirm the expected results and validity of the assay and should be performed at regular intervals.
■ Quality control tests should also be performed whenever there is any question concerning the validity of the test results.
■ Control materials are provided on demand with **AFIAS Anti-CCP Plus**. For more information regarding obtaining the control materials, contact Boditech Med Inc.’s Sales Division for assistance.
(Please refer to the instructions for use of control material.)

PERFORMANCE CHARACTERISTICS

- **Analytical sensitivity**
- Limit of Blank (LoB) 0.88 U/mL
- Limit of Detection (LoD) 3.20 U/mL

- **Analytical Specificity**
- **Cross-reactivity**
Biomolecules such as below the ones in the table were added to the test sample(s) at concentrations much higher than their normal physiological levels in the blood. **AFIAS Anti-CCP Plus** test results did not show any significant cross-reactivity with these biomolecules.

Cross-reactant	
α-SSA	α-Jo-1
α-SSB	α-Scl-70
α-Sm	α-Ribo-P
α-RNP	anti-nuclear antibody (ANA)
α-ds-DNA	

- **Interference**
Interferents listed in the following table were added to the test sample at the concentration mentioned below. **AFIAS Anti-CCP Plus** test results did not show any significant interference with these materials.

Interferents	Concentration
Hemoglobin	500 mg/dL
Bilirubin	0.2 mg/mL
Triglyceride	2,000 mg/dL
Rheumatoid factor	78 IU/mL
Human serum albumin	12 g/dL

- **Precision**
- Between-lot
One person tested three different lots of **AFIAS Anti-CCP Plus**, ten times at each concentration of the control standard.
- Between person
Three different persons tested **AFIAS Anti-CCP Plus**, ten times at each concentration of the control standard.

- Between day
One person tested **AFIAS Anti-CCP Plus** for five days; five times at each concentration of the control standard.
- Between site
One person tested **AFIAS Anti-CCP Plus** at three different sites; five times at each concentration of the control standard.

Anti-CCP [U/mL]	Between lot		Between person		Between day		Between site	
	AVG	CV(%)	AVG	CV(%)	AVG	CV(%)	AVG	CV(%)
6.25	6.12	0.7	6.22	4.3	6.11	4.7	6.03	3.8
30.00	29.46	2.2	29.97	4.5	30.26	3.8	30.30	4.6
100.00	101.34	0.9	100.78	4.5	102.40	4.3	101.24	3.7

- **Accuracy**
The accuracy was confirmed by testing with 3 different lots of **AFIAS Anti-CCP Plus**. The tests were repeated 6 times at each concentration of the control standard.

Anti-CCP [U/mL]	Lot 1	Lot 2	Lot 3	AVG	CV(%)	Bias(%)
4.15	4.23	4.06	4.14	4.14	4.3	-0.2
5.31	5.30	5.15	5.21	5.22	3.5	-1.7
16.66	16.41	16.97	16.16	16.51	4.1	-0.9
19.79	19.43	20.21	19.65	19.76	4.7	-0.1
67.02	65.86	67.74	68.02	67.21	4.1	0.3

- **Comparability**
Anti-CCP concentration of 216 clinical samples were quantified independently with **AFIAS Anti-CCP Plus (AFIAS-6)** and **comparator A** as per prescribed test procedures. Test results were compared, and results are presented in the table below.

Total (N=216)	AFIAS-6	
	Positive	Negative
comparator A	Positive	107
	Negative	9
Positive agreement rate (≥ 5 U/mL) (%)		92.2
Negative agreement rate (< 5 U/mL) (%)		97.0
Total (%)		94.4

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Note: Please refer to the table below to identify various symbols.

	Sufficient for <n> tests
	Read instruction for use
	Use by Date
	Batch code
	Catalog number
	Caution
	Manufacturer
	Authorized representative of the European Community
	In vitro diagnostic medical device
	Temperature limit
	Do not reuse
	This product fulfills the requirements of the Directive 98/79/EC on in vitro diagnostic medical devices

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